

## Qualification Pack



# Cloud Computing Assistant

QP Code: NIE/ITS/Q1201

Version: 1.0

NSQF Level: 3

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP  
Pocket, Sector-8, Dwarka  
New Delhi-110077 || email:h.sihare@nielit.gov.in

## Qualification Pack

## Contents

|                                                                                |    |
|--------------------------------------------------------------------------------|----|
| NIE/ITS/Q1201: Cloud Computing Assistant .....                                 | 3  |
| <i>Brief Job Description</i> .....                                             | 3  |
| Applicable National Occupational Standards (NOS) .....                         | 3  |
| <i>Compulsory NOS</i> .....                                                    | 3  |
| <i>Qualification Pack (QP) Parameters</i> .....                                | 3  |
| NIE/ITS/N1201: Conceptualize Operating Systems (Windows and Linux) .....       | 5  |
| NIE/ITS/N1201: Conceptualize Operating Systems (Windows and Linux). ....       | 8  |
| NIE/ITS/N1202: Introduction to Network Management .....                        | 11 |
| NIE/ITS/N1202: Introduction to Network Management. ....                        | 15 |
| NIE/ITS/N1203: Implementing Virtualization and Basics of Cloud Computing ..... | 18 |
| NIE/ITS/N1203: Implementing Virtualization and Basics of Cloud Computing. .... | 22 |
| DGT/VSQ/N0101: Employability Skills (30 Hours) .....                           | 25 |
| Assessment Guidelines and Weightage .....                                      | 30 |
| <i>Assessment Guidelines</i> .....                                             | 30 |
| <i>Assessment Weightage</i> .....                                              | 31 |
| Acronyms .....                                                                 | 32 |
| Glossary .....                                                                 | 33 |

## Qualification Pack

### NIE/ITS/Q1201: Cloud Computing Assistant

#### Brief Job Description

This course provides comprehensive understanding of cloud technologies, covering key areas such as conceptualizing operating systems (Windows and Linux), introducing network management, and implementing virtualization. This program equips students with essential skills to navigate and optimize cloud environments, making them proficient in cloud computing fundamentals and operations

#### Personal Attributes

Problem-solving skills for addressing technical challenges. Analytical thinking to optimize cloud resources effectively. Attention to detail in configuration and management. Adaptability to stay current with evolving cloud technologies. Strong communication, time management, and teamwork skills to collaborate effectively in cloud projects

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [NIE/ITS/N1201: Conceptualize Operating Systems \(Windows and Linux\).](#)
2. [NIE/ITS/N1202: Introduction to Network Management.](#)
3. [NIE/ITS/N1203: Implementing Virtualization and Basics of Cloud Computing.](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

#### Qualification Pack (QP) Parameters

|                                      |                                  |
|--------------------------------------|----------------------------------|
| <b>Sector</b>                        | Information Technology Sector    |
| <b>Sub-Sector</b>                    | Cloud Computing & Virtualization |
| <b>Occupation</b>                    | Cloud Management                 |
| <b>Country</b>                       | India                            |
| <b>NSQF Level</b>                    | 3                                |
| <b>Credits</b>                       | 10                               |
| <b>Aligned to NCO/ISCO/ISIC Code</b> | 2511.9900                        |

### Qualification Pack

|                                                           |                                                                                                                                                                                                                           |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Minimum Educational Qualification &amp; Experience</b> | 10th grade pass<br>OR<br>8th grade pass with 2 years of NTC (or NAC) after 8th Grade pass and pursuing continuous schooling in regular school with vocational subject)<br>OR<br>8th grade pass with 2 Years of experience |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                           |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                        |
| <b>Minimum Job Entry Age</b>                              | 14 Years                                                                                                                                                                                                                  |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                        |
| <b>Next Review Date</b>                                   | 30/06/2026                                                                                                                                                                                                                |
| <b>NSQC Approval Date</b>                                 | 29/03/2023                                                                                                                                                                                                                |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                       |
| <b>Reference code on NQR</b>                              | QG-03-IT-00353-2023-V1-NIELIT                                                                                                                                                                                             |
| <b>NQR Version</b>                                        | 1                                                                                                                                                                                                                         |

## Qualification Pack

### NIE/ITS/N1201: Conceptualize Operating Systems (Windows and Linux)

#### Description

In conceptualizing these operating systems, it's essential to recognize that they cater to different user needs and have different philosophies. Windows focuses on ease of use, compatibility, and broad software support, making it suitable for everyday users and businesses. Linux, with its open-source nature and flexibility, is more commonly found in server environments, embedded systems, and for users who prefer customization and the command-line interface. Understanding the strengths and weaknesses of both systems is crucial when choosing the right one for a particular use case.

#### Scope

The scope covers the following :

- Manage Operating System (Windows & Linux )

#### Elements and Performance Criteria

##### *Gaining Knowledge of how to install and operate Linux and Windows*

To be competent, the user/individual on the job must be able to:

- PC1.** Configure Network Settings and perform user and group management
- PC2.** Knowledge of various management service roles and get acquainted with the concept of remote desktop management
- PC3.** Installation of Windows and Linux Operating System
- PC4.** Creation of disk partitions and configure Active directory
- PC5.** Conceptualization of Group policy, work in own Shell and run various Windows commands
- PC6.** Basic Commands in Linux and implement administrator tasks with sudo.
- PC7.** Acquire skills pertaining to group user management, Piping and Greping and also mounting of disk.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Gaining Knowledge of how to install and operate Linux and Windows.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Gaining Knowledge of how to install and operate Linux and Windows</i>                                                    | <b>50</b>    | <b>22</b>       | -             | -          |
| <b>PC1.</b> Configure Network Settings and perform user and group management                                                | -            | -               | -             | -          |
| <b>PC2.</b> Knowledge of t various management service roles and get acquainted with the concept of remote desktop managemen | -            | -               | -             | -          |
| <b>PC3.</b> Installation of Windows and Linux Operating System                                                              | -            | -               | -             | -          |
| <b>PC4.</b> Creation of disk partitions and configure Active director                                                       | -            | -               | -             | -          |
| <b>PC5.</b> Conceptualization of s of Group policy, work in own Shell and run various Windows commands                      | -            | -               | -             | -          |
| <b>PC6.</b> Basic Commands in Linux and implement administrator tasks with sudo.                                            | -            | -               | -             | -          |
| <b>PC7.</b> Acquire skills pertaining to group user management, Piping and Greping and also mounting of disk.               | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                            | <b>50</b>    | <b>22</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                     |
|----------------------------|-----------------------------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1201                                       |
| <b>NOS Name</b>            | Conceptualize Operating Systems (Windows and Linux) |
| <b>Sector</b>              | Information Technology Sector                       |
| <b>Sub-Sector</b>          | Cloud Computing & Virtualization                    |
| <b>Occupation</b>          | Cloud Management                                    |
| <b>NSQF Level</b>          | 3                                                   |
| <b>Credits</b>             | 2                                                   |
| <b>Version</b>             | 1.0                                                 |
| <b>Last Reviewed Date</b>  | 29/03/2023                                          |
| <b>Next Review Date</b>    | 29/03/2026                                          |
| <b>NSQC Clearance Date</b> | 29/03/2023                                          |

## Qualification Pack

### NIE/ITS/N1201: Conceptualize Operating Systems (Windows and Linux).

#### Description

Operating systems are fundamental software components that manage and facilitate the interaction between hardware and software in a computer system. Windows and Linux are two prominent operating systems, each with its unique characteristics, architectures, and use cases

#### Scope

The scope covers the following :

- Able to operate operating Window & Linux Operating System

#### Elements and Performance Criteria

##### *Operating System*

To be competent, the user/individual on the job must be able to:

- PC1.** Configure Network Settings and perform user and group management
- PC2.** Knowledge of various management service roles and get acquainted with the concept of remote desktop management
- PC3.** Installation of Windows and Linux Operating System
- PC4.** Creation of disk partitions and configure Active directory
- PC5.**
  - Conceptualization of s of Group policy, work Shell and run various Windows
  - commands
- PC6.** Basic Commands in Linux and implement administrator tasks with sudo.
- PC7.** Acquire skills pertaining to group user management, Piping and Greping and also mounting of disk.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Window Operating System
- KU2.** Linux Operating System and its command



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Operating System</i>                                                                                                    | <b>15</b>    | <b>44</b>       | -             | -          |
| <b>PC1.</b> Configure Network Settings and perform user and group management                                               | -            | -               | -             | -          |
| <b>PC2.</b> Knowledge of various management service roles and get acquainted with the concept of remote desktop management | -            | -               | -             | -          |
| <b>PC3.</b> Installation of Windows and Linux Operating System                                                             | -            | -               | -             | -          |
| <b>PC4.</b> Creation of disk partitions and configure Active directory                                                     | -            | -               | -             | -          |
| <b>PC5.</b><br>• Conceptualization of s of Group policy, work Shell and run various Windows<br>• commands                  | -            | -               | -             | -          |
| <b>PC6.</b> Basic Commands in Linux and implement administrator tasks with sudo.                                           | -            | -               | -             | -          |
| <b>PC7.</b> Acquire skills pertaining to group user management, Piping and Greping and also mounting of disk.              | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                           | <b>15</b>    | <b>45</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                           |                                                            |
|---------------------------|------------------------------------------------------------|
| <b>NOS Code</b>           | NIE/ITS/N1201                                              |
| <b>NOS Name</b>           | Conceptualize Operating Systems (Windows and Linux).       |
| <b>Sector</b>             | Information Technology Sector                              |
| <b>Sub-Sector</b>         | Cloud Computing & Virtualization                           |
| <b>Occupation</b>         | Cloud Management, Cloud Management, Cloud support services |
| <b>NSQF Level</b>         | 4                                                          |
| <b>Credits</b>            | 2                                                          |
| <b>Version</b>            | 1.0                                                        |
| <b>Last Reviewed Date</b> | NA                                                         |
| <b>Next Review Date</b>   | NA                                                         |

## Qualification Pack

### NIE/ITS/N1202: Introduction to Network Management

#### Description

Network management is essential for businesses and organizations of all sizes to ensure that their networks operate efficiently, securely, and with minimal disruptions. It involves the use of network management tools and software, often collectively referred to as Network Management Systems (NMS), to simplify and automate many of these tasks. By effectively managing their networks, organizations can provide reliable services to users, protect sensitive data, and adapt to changing technology and business requirements

#### Scope

The scope covers the following :

- Introduction to Network Management courses provide foundational knowledge in overseeing and optimizing computer networks. This course equips students with skills to monitor, troubleshoot, and ensure the efficient functioning of network infrastructures. The scope of this course extends to various industries, including telecommunications, cybersecurity, and cloud computing.

#### Elements and Performance Criteria

*Classification Of Transmission Media and will have in depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches and routers*

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of OSI ,TCP/IP Model and Function of layers, learn about various types of networks
- PC2.** Knowledge about the Benefits of Network, Components of Computer Networks and also have an insight into various topological used in networking.
- PC3.** Explain various Transmission modes, Ethernet and the Classification Of Transmission Media
- PC4.** In depth idea about various networking devices like Repeaters,Hubs, Bridges, Switches and routers
- PC5.** Introduction to TCP/IP Protocol Model in Depth, Port Numbers,Common TCP& UDP Ports and also Some important TCP/IP protocols
- PC6.**
  - IP addressing, Classes, IP Addressing Scheme, Subnetting Basics alongwith How to Create Subnets, Subnet Masks, Subnetting Class
  - C Addresses, Subnetting Class B Addresses and VLSM
- PC7.**
  - Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting
  - IP Addressing

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

## Qualification Pack

- KU1.** • Classification Of Transmission Media and will have in depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches  
• and routers
- KU2.** Conceptualization of OSI ,TCP/IP Model and Function of layers, Knowledge about various types of networks

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Classification Of Transmission Media and will have in depth idea about various networking devices like Repeaters, Hubs, Bridges, Switches and routers</i>                                                                           | 50           | 23              | 15            | -          |
| <b>PC1.</b> Overview of OSI ,TCP/IP Model and Function of layers, learn about various types of networks                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> Knowledge about the Benefits of Network, Components of Computer Networks and also have an insight into various topological used in networking.                                                                             | -            | -               | -             | -          |
| <b>PC3.</b> Explain various Transmission modes, Ethernet and the Classification Of Transmission Media                                                                                                                                  | -            | -               | -             | -          |
| <b>PC4.</b> In depth idea about various networking devices like Repeaters,Hubs, Bridges, Switches and routers                                                                                                                          | -            | -               | -             | -          |
| <b>PC5.</b> Introduction to TCP/IP Protocol Model in Depth, Port Numbers,Common TCP& UDP Ports and also Some important TCP/IP protocols                                                                                                | -            | -               | -             | -          |
| <b>PC6.</b><br>• IP addressing, Classes, IP Addressing Scheme, Subnetting Basics alongwith How to Create Subnets, Subnet Masks, Subnetting Class<br>• C Addresses, Subnetting Class B Addresses and VLSM                               | -            | -               | -             | -          |
| <b>PC7.</b><br>• Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting<br>• IP Addressing | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                       | <b>50</b>    | <b>23</b>       | <b>15</b>     | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                    |
|----------------------------|------------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1202                      |
| <b>NOS Name</b>            | Introduction to Network Management |
| <b>Sector</b>              | Information Technology Sector      |
| <b>Sub-Sector</b>          | Cloud Computing & Virtualization   |
| <b>Occupation</b>          | Cloud Management                   |
| <b>NSQF Level</b>          | 3                                  |
| <b>Credits</b>             | 2.5                                |
| <b>Version</b>             | 1.0                                |
| <b>Last Reviewed Date</b>  | 29/03/2023                         |
| <b>Next Review Date</b>    | 29/03/2026                         |
| <b>NSQC Clearance Date</b> | 29/03/2023                         |

## Qualification Pack

### NIE/ITS/N1202: Introduction to Network Management.

#### Description

Network Management is the practice of administering, monitoring, and maintaining a computer network to ensure its optimal performance, security, and reliability. It involves a set of activities, tools, and techniques used to control, plan, deploy, and optimize network resources, services, and infrastructure. Effective network management is crucial for organizations of all sizes to ensure their networks function efficiently and securely.

#### Scope

The scope covers the following :

- Able to configure and manage network system

#### Elements and Performance Criteria

##### *Network Management*

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of OSI ,TCP/IP Model and Function of layers, learn about various types of networks
- PC2.** Knowledge about the Benefits of Network, Components of Computer Networks and also have an insight into various topology used in networking
- PC3.** Explain various Transmission modes, Ethernet and the Classification Of Transmission Media.
- PC4.** In-depth idea about various networking devices like Repeaters, Hubs, Bridges,Switches and router
- PC5.** Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols
- PC6.** IP addressing, Classes, IP Addressing Scheme, Subnetting Basics along with How to Create Subnets, Subnet Masks, Subnetting Class C Addresses, Subnetting Class B Addresses and VLSM
- PC7.** Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting IP Addressing

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Overview of TCP/IP & OSI Model
- KU2.** Network devices, IP Addressing & Topology configuration

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Network Management</i>                                                                                                                                                                                                    | <b>35</b>    | <b>25</b>       | -             | -          |
| <b>PC1.</b> Overview of OSI ,TCP/IP Model and Function of layers, learn about various types of networks                                                                                                                      | -            | -               | -             | -          |
| <b>PC2.</b> Knowledge about the Benefits of Network, Components of Computer Networks and also have an insight into various topology used in networking                                                                       | -            | -               | -             | -          |
| <b>PC3.</b> Explain various Transmission modes, Ethernet and the Classification Of Transmission Media.                                                                                                                       | -            | -               | -             | -          |
| <b>PC4.</b> In-depth idea about various networking devices like Repeaters, Hubs, Bridges,Switches and router                                                                                                                 | -            | -               | -             | -          |
| <b>PC5.</b> Introduction to TCP/IP Protocol Model in Depth, Port Numbers, Common TCP& UDP Ports and also Some important TCP/IP protocols                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> IP addressing, Classes, IP Addressing Scheme, Subnetting Basics along with How to Create Subnets, Subnet Masks, Subnetting Class C Addresses, Subnetting Class B Addresses and VLSM                              | -            | -               | -             | -          |
| <b>PC7.</b> Configure ip addresses in Windows and Linux and also have a proper introduction to RIR, registries and IANA along with attaining knowledge about TCP/IP Troubleshooting utilities, Troubleshooting IP Addressing | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                             | <b>35</b>    | <b>25</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                           |                                                            |
|---------------------------|------------------------------------------------------------|
| <b>NOS Code</b>           | NIE/ITS/N1202                                              |
| <b>NOS Name</b>           | Introduction to Network Management.                        |
| <b>Sector</b>             | Information Technology Sector                              |
| <b>Sub-Sector</b>         | Cloud Computing & Virtualization                           |
| <b>Occupation</b>         | Cloud Management, Cloud support services, Cloud Management |
| <b>NSQF Level</b>         | 4                                                          |
| <b>Credits</b>            | 2                                                          |
| <b>Version</b>            | 1.0                                                        |
| <b>Last Reviewed Date</b> | NA                                                         |
| <b>Next Review Date</b>   | NA                                                         |

## Qualification Pack

# NIE/ITS/N1203: Implementing Virtualization and Basics of Cloud Computing

## Description

virtualization is a foundational technology in cloud computing. Cloud providers use virtualization to create and manage the resources they offer to clients, which allows for dynamic resource allocation and efficient use of physical hardware. Understanding both virtualization and cloud computing is crucial for IT professionals and organizations looking to optimize their infrastructure and stay competitive in the digital era.

## Scope

The scope covers the following :

- This course equips professionals with the knowledge to optimize IT infrastructure through virtualization and harness the benefits of cloud computing for enhanced business agility. Gain expertise in creating virtual environments for efficient resource utilization and workload management. Understanding the fundamental concepts of cloud services, scalability, and on-demand resource provisioning. Exploring virtualization technologies like VMware and Hyper-V, enhancing skills crucial for modern IT infrastructures. Cloud computing fundamentals cover key aspects such as infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS).

## Elements and Performance Criteria

### *Differentiate and benefit of Open source Virtualisation and proprietary Virtualisation*

To be competent, the user/individual on the job must be able to:

**PC1.** Differentiating between Open source Virtualization and proprietary Virtualisation

**PC2.** Installation and use Open source virtualization tools - KVM, Hyper-V and Xen

**PC3.** Creation of Virtual Machine using Virtualization Tools

### *Conceptualize Cloud Characteristics along with Features Of Cloud Computing.*

To be competent, the user/individual on the job must be able to:

**PC4.** Conceptualize Cloud Characteristics and Working of Cloud Computing along with Features Of Cloud Computing

**PC5.** Knowledge pertaining to Types of Cloud

**PC6.** Enlightened about the Benefits of Cloud Computing and Applications of Cloud Computing In industries

**PC7.** In-depth coverage of Cloud Platform ranging from its benefits to its implementation and its scope and its pros and cons

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



## Qualification Pack

- KU1.** Differentiate and benefit of Open source Virtualization and proprietary Virtualisation
- KU2.** Conceptualize Cloud Characteristics along with Features Of Cloud Computing

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Differentiate and benefit of Open source Virtualisation and proprietary Virtualisation</i>                                       | <b>50</b>    | <b>23</b>       | <b>10</b>     | -          |
| <b>PC1.</b> Differentiating between Open source Virtualization and proprietary Virtualisation                                       | -            | -               | -             | -          |
| <b>PC2.</b> Installation and use Open source virtualization tools - KVM, Hyper-V and Xen                                            | -            | -               | -             | -          |
| <b>PC3.</b> Creation of Virtual Machine using Virtualization Tools                                                                  | -            | -               | -             | -          |
| <i>Conceptualize Cloud Characteristics along with Features Of Cloud Computing.</i>                                                  | <b>50</b>    | <b>22</b>       | <b>5</b>      | -          |
| <b>PC4.</b> Conceptualize Cloud Characteristics and Working of Cloud Computing along with Features Of Cloud Computing               | -            | -               | -             | -          |
| <b>PC5.</b> Knowledge pertaining to Types of Cloud                                                                                  | -            | -               | -             | -          |
| <b>PC6.</b> Enlightened about the Benefits of Cloud Computing and Applications of Cloud Computing In industries                     | -            | -               | -             | -          |
| <b>PC7.</b> In-depth coverage of Cloud Platform ranging from its benefits to its implementation and its scope and its pros and cons | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                    | <b>100</b>   | <b>45</b>       | <b>15</b>     | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                           |
|----------------------------|-----------------------------------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1203                                             |
| <b>NOS Name</b>            | Implementing Virtualization and Basics of Cloud Computing |
| <b>Sector</b>              | Information Technology Sector                             |
| <b>Sub-Sector</b>          | Cloud Computing & Virtualization                          |
| <b>Occupation</b>          | Cloud Management, Cloud Management                        |
| <b>NSQF Level</b>          | 3                                                         |
| <b>Credits</b>             | 4.5                                                       |
| <b>Version</b>             | 1.0                                                       |
| <b>Last Reviewed Date</b>  | 29/03/2023                                                |
| <b>Next Review Date</b>    | 29/03/2026                                                |
| <b>NSQC Clearance Date</b> | 29/03/2023                                                |

## Qualification Pack

# NIE/ITS/N1203: Implementing Virtualization and Basics of Cloud Computing.

## Description

Virtualization and Cloud Computing are two closely related concepts in modern IT infrastructure. They offer numerous benefits, including resource optimization, cost reduction, scalability, and flexibility. Here's an overview of each concept and how they are implemented:

## Scope

The scope covers the following :

- Able to understand Cloud computing and its Virtualization methodology

## Elements and Performance Criteria

### Virtualization

To be competent, the user/individual on the job must be able to:

- PC1.** Differentiating between Open source Virtualization and proprietary Virtualization.
- PC2.** Installation and use Open source virtualization tools - KVM, Hyper-V and Xen
- PC3.** Creation of Virtual Machine using Virtualization Tools
- PC4.** Cloud Computing Concepts and would be able to differentiate Cloud computing vs Cluster computing vs. Grid computing.
- PC5.** Conceptualize Cloud Characteristics and Working of Cloud Computing along with features Of Cloud Computing
- PC6.** Knowledge pertaining to Types of Cloud
- PC7.** Enlightened about the Benefits of Cloud Computing and Applications of Cloud Computing In industries
- PC8.** In-depth coverage of Cloud Platform ranging from its benefits to its implementation and its scope and its pros and cons

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Virtualization and its type
- KU2.** Open source Virtualization and proprietary Virtualization.
- KU3.** Creation of Virtual Machines
- KU4.** Types of Cloud & its service model

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Virtulization</i>                                                                                                                | <b>60</b>    | <b>60</b>       | -             | -          |
| <b>PC1.</b> Differentiating between Open source Virtualization and proprietary Virtualization.                                      | -            | -               | -             | -          |
| <b>PC2.</b> Installation and use Open source virtualization tools - KVM, Hyper-V and Xen                                            | -            | -               | -             | -          |
| <b>PC3.</b> Creation of Virtual Machine using Virtualization Tools                                                                  | -            | -               | -             | -          |
| <b>PC4.</b> Cloud Computing Concepts and would be able to differentiate Cloud computing vs Cluster computing vs. Grid computing.    | -            | -               | -             | -          |
| <b>PC5.</b> Conceptualize Cloud Characteristics and Working of Cloud Computing along with features Of Cloud Computing               | -            | -               | -             | -          |
| <b>PC6.</b> Knowledge pertaining to Types of Cloud                                                                                  | -            | -               | -             | -          |
| <b>PC7.</b> Enlightened about the Benefits of Cloud Computing and Applications of Cloud Computing In industries                     | -            | -               | -             | -          |
| <b>PC8.</b> In-depth coverage of Cloud Platform ranging from its benefits to its implementation and its scope and its pros and cons | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                    | <b>60</b>    | <b>60</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                           |                                                            |
|---------------------------|------------------------------------------------------------|
| <b>NOS Code</b>           | NIE/ITS/N1203                                              |
| <b>NOS Name</b>           | Implementing Virtualization and Basics of Cloud Computing. |
| <b>Sector</b>             | Information Technology Sector                              |
| <b>Sub-Sector</b>         | Cloud Computing & Virtualization                           |
| <b>Occupation</b>         | Cloud Management, Cloud Management, Cloud support services |
| <b>NSQF Level</b>         | 4                                                          |
| <b>Credits</b>            | 5                                                          |
| <b>Version</b>            | 1.0                                                        |
| <b>Last Reviewed Date</b> | NA                                                         |
| <b>Next Review Date</b>   | NA                                                         |



## Qualification Pack

### DGT/VSQ/N0101: Employability Skills (30 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

**PC1.** understand the significance of employability skills in meeting the job requirements

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

**PC2.** identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

**PC3.** explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

**PC4.** speak with others using some basic English phrases or sentences

##### *Communication Skills*

To be competent, the user/individual on the job must be able to:

**PC5.** follow good manners while communicating with others

**PC6.** work with others in a team

## Qualification Pack

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

**PC7.** communicate and behave appropriately with all genders and PwD

**PC8.** report any issues related to sexual harassment

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

**PC9.** use various financial products and services safely and securely

**PC10.** calculate income, expenses, savings etc.

**PC11.** approach the concerned authorities for any exploitation as per legal rights and laws

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

**PC12.** operate digital devices and use its features and applications securely and safely

**PC13.** use internet and social media platforms securely and safely

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

**PC14.** identify and assess opportunities for potential business

**PC15.** identify sources for arranging money and associated financial and legal challenges

### *Customer Service*

To be competent, the user/individual on the job must be able to:

**PC16.** identify different types of customers

**PC17.** identify customer needs and address them appropriately

**PC18.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC19.** create a basic biodata

**PC20.** search for suitable jobs and apply

**PC21.** identify and register apprenticeship opportunities as per requirement

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use basic spoken English language

**KU6.** Do and dont of effective communication

**KU7.** inclusivity and its importance

**KU8.** different types of disabilities and appropriate communication and behaviour towards PwD

**KU9.** different types of financial products and services

## Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the job requirements                                                                                                                                                                | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                     | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC2.</b> identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                             | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC3.</b> explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc. | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC4.</b> speak with others using some basic English phrases or sentences                                                                                                                                                                                    | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC5.</b> follow good manners while communicating with others                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC6.</b> work with others in a team                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                                                                                                               | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC7.</b> communicate and behave appropriately with all genders and PwD                                                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC8.</b> report any issues related to sexual harassment                                                                                                                                                                                                     | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                                                                                                            | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use various financial products and services safely and securely                                                                                                                                                                                    | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> calculate income, expenses, savings etc.                                             | -            | -               | -             | -          |
| <b>PC11.</b> approach the concerned authorities for any exploitation as per legal rights and laws | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                   | <b>4</b>     | <b>6</b>        | -             | -          |
| <b>PC12.</b> operate digital devices and use its features and applications securely and safely    | -            | -               | -             | -          |
| <b>PC13.</b> use internet and social media platforms securely and safely                          | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                           | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC14.</b> identify and assess opportunities for potential business                             | -            | -               | -             | -          |
| <b>PC15.</b> identify sources for arranging money and associated financial and legal challenges   | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC16.</b> identify different types of customers                                                | -            | -               | -             | -          |
| <b>PC17.</b> identify customer needs and address them appropriately                               | -            | -               | -             | -          |
| <b>PC18.</b> follow appropriate hygiene and grooming standards                                    | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC19.</b> create a basic biodata                                                               | -            | -               | -             | -          |
| <b>PC20.</b> search for suitable jobs and apply                                                   | -            | -               | -             | -          |
| <b>PC21.</b> identify and register apprenticeship opportunities as per requirement                | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                  | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0101                   |
| <b>NOS Name</b>            | Employability Skills (30 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 2                               |
| <b>Credits</b>             | 1                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 12/03/2026                      |
| <b>Next Review Date</b>    | 12/03/2029                      |
| <b>NSQC Clearance Date</b> | 12/03/2026                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.

### Qualification Pack

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only

### Minimum Aggregate Passing % at QP Level : 50

**(Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Assessment Weightage

Compulsory NOS

| National Occupational Standards                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|--------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| NIE/ITS/N1201.Conceptualize Operating Systems (Windows and Linux).       | 15           | 45              | 0             | 0          | 60          | 20         |
| NIE/ITS/N1202.Introduction to Network Management.                        | 35           | 25              | 0             | 0          | 60          | 24         |
| NIE/ITS/N1203.Implementing Virtualization and Basics of Cloud Computing. | 60           | 60              | 0             | 0          | 120         | 47         |
| DGT/VSQ/N0101.Employability Skills (30 Hours)                            | 20           | 30              | -             | -          | 50          | 9          |
| <b>Total</b>                                                             | <b>130</b>   | <b>160</b>      | <b>0</b>      | <b>0</b>   | <b>290</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |



## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |